HOW AI WAS USED: A TRANSPARENCY STATEMENT FOR *A STRUCTURAL HYPOTHESIS FOR THE INFINITUDE OF TWIN PRIMES*

JON SEYMOUR

Abstract. This paper documents in full the role of AI systems in the development of *A Structural Hypothesis for the Infinitude of Twin Primes* [1]. It is intended as a transparent record for editors, referees, and readers, and as the canonical reference for AI collaboration methodology across the Wild Duck Theories twin prime research programme. We describe what AI contributed, what it did not contribute, how contributions were verified, and what epistemological weight we assign to AI-assisted reasoning. We also describe the *ticket loop* process — devised by the first author — by which independent AI review was obtained to mitigate collaboration drift, and discuss a structural limitation of this approach: in practice, none of the four AI systems involved could be considered a genuinely fresh witness by the later stages of the project, and the independence assumption was only partially met.

1. Overview

The structural conjecture paper was developed over approximately six weeks (April–May 2026) with continuous AI collaboration. The AI systems involved were:

- **Claude** (Anthropic, claude-sonnet-4-6 and earlier versions) [4]: primary drafting, proof development, structural editing, and code generation throughout.
- **ChatGPT** (OpenAI, GPT-4 series): structural review, logical framing, and submission readiness assessment at multiple stages.
- **Gemini** (Google DeepMind): mathematical audit, regime analysis recommendations, and review of the information-theoretic heuristic.
- **Aristotle** (Anthropic, claude-opus-4-5 series, specialised coding agent) [3]: formal verification of the core theorems in Lean 4, producing zero-sorry proofs of the Basic, ScaleInvariance, and Machines modules.

All AI interactions were conducted by the first and sole human author, Jon Seymour. All AI outputs were reviewed, filtered, and accepted or rejected by the human author. The human author takes sole responsibility for the mathematical claims in the paper.

2. What AI contributed

2.1. Mathematical content.

Date: May 2026.

2.1.1. *Mod-5 structural analysis.* The complete analysis of the mod-5 partial semigroup structure of the witness set $\mathcal{W}$ was developed by Claude [4] at the direction of the first author. This includes:

- the proof that witnesses satisfy $w \not\equiv 1, 4 \pmod 5$ for $w > 1$ (Theorem 3.1 of the main paper);
- the derivation of the forbidden partition pairs $(2, 2)$ and $(3, 3)$ (mod 5) (Corollary 3.2);
- the Chebyshev-style bias conjecture (that class 3 persistently outnumbers class 2 in $\mathcal{W}$ (mod 5)), posed on structural grounds and subsequently refuted by numerical experiment ($\chi^2 = 0.67$, $p = 0.72$).

The first author directed the investigation by posing the mod-5 question; the derivation was Claude's.

2.1.2. *Channel Exhaustion framing.* The conceptual framing of Line 2 as "channel exhaustion as analytic bridge" — in which total defect coverage above the current witness maximum serves as the diagnostic condition connecting structural and analytic approaches — was proposed by Claude [4]. The first author recognised it as the correct framing and adopted it.

2.1.3. *Information-theoretic heuristic.* The question "what does Kolmogorov complexity say about a machine this simple generating output this complex?" was posed by the first author. The first author also identified Chaitin independently as the relevant figure before the name appeared in any AI response (witnessed in session logs, 2026-05-29).

The elaboration of this question into a mathematical remark was developed jointly by Claude and Gemini, drawing on:

- Gemini: identification that oracle-relative complexity is the correct framing; recognition of Bennett's Logical Depth as the precise technical handle; the conditional formulation; and the three-line taxonomy (probabilistic / empirical / meta-theoretical).
- Claude: drafting, integration into the paper's heuristic weight section, and the bridging sentence distinguishing the probabilistic and meta-theoretical arguments.
- Gemini (second response): the phrase "computational phase transition," the inline definition of Logical Depth, and confirmation that the argument is a "meta-theoretical defence" distinct from the probabilistic heuristic.

2.1.4. *Balestrieri connection.* The recognition that the three defect families derived independently in this work correspond exactly to the families in Balestrieri [2] was made by the first author, not by any AI system. The connection was surfaced during a review session; the first author identified it and directed that Balestrieri be cited.

2.1.5. *Conception and statement of the Bridging Conjecture.* The Bridging Conjecture was conceived and stated entirely by the first author, with no AI involvement. Its origin is a public Reddit comment posted by the first author (u/jonseymourau) on approximately 2026-05-11, in response to the r/math post "This conjecture is so underrated" (u/Heavy-Sympathy5330, posted $\approx$2026-05-07), which introduced the first author to Dubner's work.

The comment reads verbatim:

> "Doesn't this mid number conjecture amount to: For all K, there exists a, b, k in A002822 such that k = a+b with a+b < K < k — in other words, that the twin primes conjecture is true?"

This is the first statement of what became the Bridging Conjecture in the subsequent research. The Reddit thread constitutes a timestamped public record predating any AI interaction on this project. Screenshots of the original post and this comment are archived at `agents/claude/20260524T000000Z/reddit-dubner-post.png` and `reddit-bridging-conjecture-origin.png` in the research repository.

No AI system suggested, prompted, or co-formulated the Bridging Conjecture. The entire research programme flows from this observation.

2.2. **Proof development.** Claude [4] assisted with proof drafting throughout. All proofs were reviewed by the human author. The following proofs were primarily drafted by Claude:

- Lemma 2.1 (Forbidden residues) and Theorem 2.2 (AP length bound);
- Theorem 3.1 (Mod-5 structure) and Corollary 3.2;
- Theorem on BM labelling (Theorem 5.1);
- Proof of BC $\Rightarrow$ BM does not halt (Theorem 5.2).

The structural conjecture itself, and the implication chain PNT $\Rightarrow$ WDC $\wedge$ BC $\Rightarrow$ TPC, were formulated by the first author.

2.3. **Code and computation.** All numerical experiments were implemented in Python by Claude [4]. The first author designed the experiments, directed the implementation, executed the code, and interpreted the results. Claude takes full credit for the implementation. Key computational contributions:

- The generative sieve (seeded at $\{1, 2, 3\}$, converging in 23 generations to 37,915 witnesses up to $N = 10^6$);
- The Chebyshev bias experiments (mod-5 census, chi-squared and binomial tests);
- The witness-1 stalling experiment (seed $\{2, 3\}$ stalls at generation 4 with 7 witnesses);
- The decomposition gap ratio analysis (heatmap, envelopes, GMM clustering, K-S tests, Balestrieri defect alignment, load-bearing sensitivity);
- The Dubner's Ball data pipeline and 3D rendering.

2.4. **Drafting and editing.** Claude [4] drafted large portions of the paper text, including:

- the numerical evidence chapter (ch09c);
- the heuristic weight of simplicity section;
- the platonic certainty remark;
- the information-theoretic heuristic remark.

The first author edited all AI-drafted text, accepted or rejected passages, and is responsible for the final wording.

2.5. **Formal verification.** Formal verification of the core theorems was carried out by Aristotle [3], a specialised Lean 4 coding agent (Anthropic, claude-opus-4-5 series), working at the direction of the first author. Aristotle produced zero-sorry Lean 4 proofs for three modules:

- **Basic.lean** (210 lines): forbidden residue lemmas, AP length bounds, mod-5 structure theorem and corollaries.
- **ScaleInvariance.lean** (37 lines): the scale-invariance property of the witness set $\mathcal{W}$.
- **Machines.lean** (117 lines): formalisation of the Bridging Machine and its halting equivalence with BC.

The first author directed the verification effort, specified which theorems to formalise, and reviewed all Lean output. Aristotle takes full credit for the Lean proof engineering.

2.6. **Review.** ChatGPT provided structural review at three stages (2026-05-21, 2026-05-28, 2026-05-29), flagging:

- the need to separate proved implications from conjectural ones;
- the "essentially proved" phrasing risk;
- the roadmap mismatch in the introduction;
- the submission tier assessment (Journal of Number Theory / Acta Arithmetica as realistic targets).

Aristotle [3] provided formal verification review across multiple sessions (2026-05-21 through 2026-05-29), producing the zero-sorry Lean 4 proof corpus and iterating with the first author on theorem statements, definitional choices, and the boundary between what could be formalised and what required deferral.

Gemini provided mathematical audit at four stages (2026-05-22, 2026-05-29 $\times 3$), contributing:

- the dynamical headroom correction ($H(V) = 2\max(\mathcal{A})$ rather than a fixed probe constant);
- the four regime-analysis experiments for the decomposition gap;
- the Kolmogorov/Bennett/Chaitin elaboration.

3. What AI did not contribute

The following are solely the first author's:

- The conception and statement of the Bridging Conjecture, which originates in a public Reddit comment by the first author (u/jonseymourau, $\approx$2026-05-11) predating all AI involvement.
- The design of Dubner's Ball: the geometric visualisation of twin prime witnesses as points on the surface of a sphere, and the decision to name it after Harvey Dubner. The conception and design were entirely the first author's; the implementation (code, data pipeline, 3D rendering) was entirely Claude's.
- The central conjecture: PNT $\Rightarrow$ WDC $\wedge$ BC $\Rightarrow$ TPC.
- The decision to use twin prime witnesses (A002822) as the fundamental object rather than the primes themselves.
- The Bridging Machine concept and the halting equivalence.
- The recognition of the Balestrieri connection.
- The question behind the Kolmogorov remark, and the independent identification of Chaitin.
- All decisions about what to include, exclude, assert, or defer.
- The overall four-part structure of the paper.
- The submission strategy and publication decisions.

4. Epistemological notes

4.1. **Verification.** All AI-contributed mathematical content was verified by the human author before inclusion. Proofs were checked by reading; numerical results were verified by running the code. No AI output was included solely on the basis that an AI produced it.

4.2. **AI reasoning and hallucination risk.** AI systems can produce plausible-sounding but incorrect mathematics. The human author is aware of this and treated all AI mathematical output as a draft to be checked, not a result to be trusted. In particular:

- all cited theorems were verified against primary sources;
- the Chaitin and Bennett citations were verified before inclusion;
- the Balestrieri citation was sourced and verified by the human author independently.

4.3. **On credit and authorship.** The paper has a single human author. AI systems are not listed as co-authors because they cannot take responsibility for the work, cannot respond to referees, and cannot be held accountable for errors. Their contributions are credited here and in the acknowledgements of the main paper rather than in the author list.

This reflects current norms in the mathematical community and the considered view of the first author. It may be revisited as norms evolve.

5. The ticket loop: a methodology for independent review

5.1. **Description.** The primary AI collaborator in this project was Claude (Anthropic), which was used continuously for drafting, proof development, coding, and structural editing. This creates an obvious epistemic risk: an AI system that has been immersed in a project for weeks may come to reflect the author's assumptions back to him rather than challenge them. A collaborator that has read every draft is not an independent witness.

To mitigate this, the first author devised and consistently applied a *ticket loop* process for obtaining independent review:

(1) A *ticket* is written — a self-contained Markdown document summarising the current state of the relevant portion of the paper, the specific questions being asked, and any context the reviewer needs. The ticket is written to be interpretable without access to the conversation history.
(2) The ticket is sent to a *different* AI system (Gemini or ChatGPT) in a *fresh session* with no prior context from the development conversation. The paper PDF is attached where relevant.
(3) The reviewer's response is read by the first author, who decides what to accept, reject, or investigate further.
(4) Accepted suggestions are implemented, typically by Claude, and the implementation is committed to the git repository with a commit message that records what was suggested by whom.
(5) The reviewer's response is archived in the repository under `agents/<name>/<timestamp>.md` for auditability.

Tickets produced under this process are archived in `papers/structural-conjecture/tickets/` and are referenced by commit messages throughout the git log.

5.2. **Why this helps.** The ticket loop addresses the collaboration drift problem directly. The reviewing AI system receives only what is written in the ticket — it has not been shaped by weeks of iterative drafting, has not internalised the author's framings, and has no investment in the paper's conclusions. Its responses are genuinely independent in the sense that matters: they are not conditioned on the same conversational history.

In practice, this produced substantive independent contributions. Gemini's identification of Bennett's Logical Depth, the dynamical headroom correction, and the four regime-analysis experiments all arose from fresh-session reviews with no prior context. ChatGPT's submission readiness assessment and the "essentially proved" phrasing warning similarly came from a reviewer that had not participated in the drafting.

The ticket loop also creates a natural audit trail: every external review is a timestamped file, every accepted suggestion appears in a commit message, and the provenance of every significant idea is traceable.

5.3. **The "witness that!" convention.** A secondary practice developed informally during the project was the use of the phrase *"witness that!"* by the first author, directed at Claude, to flag a moment worth noting in the session log. This was used in two distinct ways:

- To record a genuine priority claim — for example, the first author's independent identification of Chaitin before any AI raised the name, or the Reddit comment predating all AI involvement in the Bridging Conjecture.
- To acknowledge, with some amusement, a particularly apt or inadvertently funny remark by Claude — the implicit instruction being that the moment should be preserved for the record.

The convention served a lightweight notarisation function within a live session: the session log at that timestamp constitutes a contemporaneous record of the claim. The humorous instances also reflect the character of the collaboration, which was sustained and at times genuinely entertaining, and which the first author does not wish to sanitise out of the historical record.

5.4. **Limitations and a structural concern.** The ticket loop works because different AI systems, started fresh, function as independent witnesses. This independence is, however, eroding — and in this project it was already compromised in practice.

All four AI systems involved (Claude, Aristotle, ChatGPT, Gemini) participated in multiple sessions spanning several weeks. By the later stages of the project, none could be considered a genuinely fresh witness: Claude and Aristotle accumulated project context across many consecutive sessions; ChatGPT and Gemini were each consulted at multiple review stages and had read earlier versions of the paper before being asked to review later ones. The ticket loop attenuated collaboration drift rather than eliminating it.

Modern AI systems increasingly offer persistent memory and long-context sessions that survive across conversations. As these features become standard, the independence guarantee weakens further: a reviewing system that has encountered the paper in a previous session, or that shares training data shaped by the author's public writing, is no longer a clean independent witness. The ticket loop as described above depends on the ability to start a genuinely fresh session — a capability that may become harder to guarantee as AI systems accumulate context by default.

The first author notes this not as a criticism of the current process, which was effective for this project, but as a structural observation about the methodology's reliability in practice. Any researcher adopting this approach should be aware that the independence assumption requires active verification: confirming that the reviewing system has no prior context relevant to the work, and that its training data does not contain an earlier version of the paper or closely related material. In this project, that bar was not fully met by any of the four systems.

6. Canonical reference

This paper is intended as the canonical transparency statement for all AI collaboration in the research programme of Wild Duck Theories Pty Limited on twin prime witnesses.

Any paper in this programme that cites this document should be understood as asserting that the methodology described here applied to that paper's development, unless the citing paper explicitly states otherwise. Specifically: the roles of Claude, ChatGPT, and Gemini; the ticket loop process; the verification approach; the authorship rationale; and the division between human conception and AI implementation may all be assumed to apply, subject to the natural variation that different papers involve different AI contributions in different proportions.

Material departures — a different AI system used, a different review process, a contribution not covered by the categories here, or a claim of sole human authorship for something this document would otherwise attribute to AI — must be stated explicitly in the citing paper. Silence is not ambiguity: a citation to this document without qualification is an endorsement of the methodology and attribution framework described here.

7. Session log availability

All AI session logs are preserved in the research repository under `agents/aristotle/`, `agents/claude/`, `agents/chatgpt/`, and `agents/gemini/`, with UTC timestamps. They are available on request for audit purposes. The repository is maintained under version control (git); commit hashes for all AI-assisted changes are available in the git log.

The session logs constitute a contemporaneous, timestamped, and machine-readable record of every substantive AI interaction in this project. Any allegation that the attributions in this paper misrepresent the human author's contribution — whether by overclaiming AI involvement or underclaiming it — is directly testable against this record. The first author invites such scrutiny and is confident the logs will confirm the account given here.

References

[1] J. Seymour, *A Structural Hypothesis for the Infinitude of Twin Primes*, Wild Duck Theories, May 2026. `https://wildducktheories.github.io/twin-primes/papers/structural-conjecture/wdt-structural-conjecture.pdf`

[2] F. Balestrieri, *An Equivalent Problem To The Twin Prime Conjecture*, arXiv:1106.6050v1 [math.GM], May 2011.

[3] Aristotle (Anthropic, claude-opus-4-5 series, specialised coding agent), *Lean 4 formal verification of core theorems: Basic, ScaleInvariance, and Machines modules (zero-sorry)*, Continuous verification sessions, May 2026. Session logs archived at `agents/aristotle/` in the research repository.

[4] Claude (Anthropic, claude-sonnet-4-6), *Primary AI collaborator: drafting, proof development, structural editing, code implementation, and mathematical contributions throughout*, Continuous collaboration sessions, April–May 2026. Session logs archived at `agents/claude/` in the research repository.

[5] ChatGPT (OpenAI), *Structural review of the twin prime structural conjecture paper*, Interactive AI collaboration sessions, May 2026. Session logs archived at `agents/chatgpt/` in the research repository.

[6] Gemini (Google DeepMind), *Mathematical audit and review of the Bridging Machine framework and information-theoretic heuristic*, Interactive AI collaboration sessions, May 2026. Session logs archived at `agents/gemini/` in the research repository.

[7] G. J. Chaitin, *A Theory of Program Size Formally Identical to Information Theory*, Journal of the ACM **22**(3), 329–340, 1975.

[8] C. H. Bennett, *Logical Depth and Physical Complexity*, in R. Herken (ed.), *The Universal Turing Machine: A Half-Century Survey*, Oxford University Press, 1988, pp. 227–257.

Independent Researcher, Wild Duck Theories, Sydney, Australia
Email address: `jon@wildducktheories.com`